

The Illusion of Equivalence: Systematic FP16 Divergence in KV-Cached Autoregressive Inference

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Abstract

KV caching is deployed in virtually every production autoregressive transformer inference system under an assumption of numerical equivalence to cache-free computation that is rarely tested. We show this assumption fails under standard FP16 inference and, unlike prior nondeterminism studies that characterize divergence as an emergent statistical property of system configuration, localize it to a specific locus—the KV cache state—confirmed both by elimination (residual-stream patching) and by direct positive confirmation (KV-cache tensor patching)—and to an architectural amplification mechanism, the Grouped-Query Attention (GQA) sharing ratio, that has not previously been connected to numerical divergence. Across three open-weight models (LLaMA-2-7B, Mistral-7B-v0.3, Gemma-2-2B) evaluated on GSM8K, cache-ON and cache-OFF execution paths produce deterministically divergent token sequences in 100% of runs, including under greedy decoding. A controlled FP32 falsification drops the flip rate to exactly 0.0% and reduces KL divergence by more than eight orders of magnitude, isolating FP16 non-associativity as the sole cause. Layer-wise drift profiling reveals architecturally predictable propagation patterns; activation patching of the residual stream fails to recover the cache-free trajectory, while direct KV-tensor patching recovers cache-ON’s output exactly. We further show that 98.4% of token-divergent examples produce different final answers, that the divergence persists at comparable severity under BF16, and that precision elevation confined to the attention computation alone is insufficient to mitigate the effect.

1 Introduction

Autoregressive transformer inference is almost universally based on the Key-Value (KV) cache: instead of recomputing attention over the entire prefix at every decoding step, previously computed

keys and values are stored and retrieved. This technique, found in practically every production serving system (Kwon et al., 2023), efficient attention kernel (Dao et al., 2022; Dao, 2023), and prompt-caching API, is typically assumed to be *numerically equivalent* to cache-free inference. This assumption is rarely tested directly because KV caching is treated as a pure performance optimization rather than a numerically consequential design choice.

The failure. Cache-ON and cache-OFF execution paths use different kernel layouts and, as a direct consequence, different floating-point accumulation orders under the non-associativity of FP16 arithmetic. This structural difference produces systematic numerical divergence in the KV tensors written during decoding. Unlike the stochastic variability demonstrated in previous work (Zheng et al., 2023; Chen et al., 2023), this divergence is *deterministic and reproducible*: for the same input, model, and hardware/software environment, the cache-ON and cache-OFF inference paths always produce different output tokens. This determinism is central to our characterization; the divergence is not statistical variance that could average out with more data, but a specific, reproducible property of FP16 cached inference, confirmed by a controlled FP32 falsification (Section 5.3) that eliminates it entirely rather than merely reducing it.

Relation to prior work. Numerical precision research (Micikevicius et al., 2017) addresses training stability and model compression (Frantar et al., 2022; Dettmers et al., 2022), not cache-ON/cache-OFF equivalence at inference time. KV cache optimization work (Kwon et al., 2023; Shazeer, 2019; Ainslie et al., 2023) treats caching as a memory/throughput problem and assumes numerical equivalence without verifying it. Mechanistic interpretability methods (Meng et al., 2022; Wang et al., 2022) intervene on the residual stream but have not

considered the stateful KV cache itself as a locus of divergence. Most closely related, concurrent work (Yuan et al., 2025) characterizes FP16/BF16 non-associativity across batch size, GPU count, and hardware configuration. We isolate a structurally distinct comparison—cache-ON versus cache-OFF execution on *identical* hardware and configuration—and further localize the effect to the KV cache state and characterize its amplification via the GQA sharing ratio, neither of which prior work addresses. A mitigation such as LayerCast, designed for cross-configuration nondeterminism, would not be expected to resolve cache-ON/cache-OFF divergence on fixed hardware because our divergence has a different mechanism and a different causal locus (Section 5.5).

This paper. We provide a systematic empirical analysis across three open-weight language models—LLaMA-2-7B, Mistral-7B-v0.3, and Gemma-2-2B—evaluated on GSM8K (Cobbe et al., 2021) under greedy, top- k , and top- p sampling. Our analysis proceeds through a causal chain: we first establish that cache-ON/cache-OFF consistently diverge, then characterize layer-by-layer propagation, identify FP16 accumulation order as the root cause, resolve an apparent contradiction between flip index and KL divergence, and localize divergence to the KV cache state. Additional experiments isolate the pure numerical effect from context-compounding, test precision robustness, and quantify practical outcome impact.

Contributions. Our results point to a single mechanistic interpretation: FP16 rounding errors are written into KV tensors at each decode step, accumulate across layers and sequence positions, and cannot be corrected by downstream residual-stream interventions. Grouped-Query Attention (GQA (Ainslie et al., 2023)) broadcasts a single FP16 rounding error to all attending query heads simultaneously, amplifying divergence relative to standard Multi-Head Attention (MHA (Vaswani et al., 2017)). Specifically, our findings include:

(i) **Behavioral characterization.** Token divergence occurs in 100% of runs, even under greedy decoding, ruling out sampling as a cause. Cache-ON yields higher accuracy in 8 of 9 conditions (McNemar, Bonferroni-corrected $p < 0.05$), the signature of a systematic execution-path bias.

(ii) **Mechanistic root cause.** Switching to FP32 reduces KL divergence by over eight orders

of magnitude (to 2.5×10^{-8}) and eliminates token flips entirely (flip rate 0.0%). The accumulation error is flat across sequence lengths ($\approx 0.036\%$), indicating it is determined by model architecture rather than input content or length.

(iii) **Decision boundary resolution.** Examples with earlier flip steps also show higher mean KL—an apparent contradiction with an accumulation account. We show both are joint downstream consequences of decision-boundary proximity (low logit margin), not causally linked to each other (Welch t -test, $p < 0.05$ in 8 of 9 conditions).

(iv) **Causal localization.** Residual-stream patching retrieves a median of $<1\%$ of KV divergence; patching all layers jointly yields *negative* mean recovery, pinning the causal variable to the KV cache state.

(v) **Direct causal confirmation.** Patching cache-ON’s key/value tensors into cache-OFF’s cache at decode step 0 recovers cache-ON’s output exactly (100% at full-layer patch). Per-layer recovery dissociates sharply from passive drift: Mistral’s layer 1 amplifies drift by $35\times$ but its single-layer patch recovers under 1% of divergence; LLaMA-2’s largest drift layer accounts for 39% of recovery on its own.

(vi) **Isolating the pure numerical effect.** Teacher-forcing cache-OFF onto cache-ON’s realized token sequence at every step shows that 80–93% of the naively observed KL reflects context-compounding rather than the pure numerical perturbation, which nevertheless remains clearly non-zero and orders of magnitude above the FP32 floor.

(vii) **Precision robustness and locus refinement.** The divergence persists at comparable severity under BF16. Elevating only the attention computation to FP32 provides no measurable mitigation, indicating that sustained full-pass precision is required.

(viii) **Practical impact.** Among token-divergent examples, 98.4% produce a different final extracted answer; cross-cache-mode reconvergence is lower than under ordinary sampling variance alone.

2 Background and Related Work

KV cache, execution paths, and floating-point accumulation. The KV cache stores keys and values from previously computed attention stages so that cache-ON inference can skip prefix recomputation. Cache-ON and cache-OFF inference are mathematically equivalent under *exact* arithmetic but use different kernel layouts and accumulation orders under finite precision. Both GQA (Ainslie et al., 2023) and MQA (Shazeer, 2019) share KV heads across query groups, altering the reduction structure in ways that change how FP16 rounding errors propagate; KV caching makes these discrepancies *stateful*, persisting across decode steps.

Floating-point arithmetic is not associative (Goldberg, 1991):

$$(a+b)+c \neq a+(b+c), \quad (a \times b) \times c \neq a \times (b \times c) \quad (1)$$

for finite-precision values a, b, c , even though both identities hold exactly for real numbers. Since attention computation is dominated by large dot-product accumulations, it is *addition* non-associativity that drives the divergence reported here; multiplication non-associativity plays a secondary role. Low-precision formats such as FP16 (10-bit mantissa) and BF16 (7-bit mantissa) are more sensitive to accumulation order than FP32. Previous research has examined reduced precision and quantization (Micikevicius et al., 2017; Xiao et al., 2023b) largely for efficiency rather than inference correctness. Concretely, cache-OFF computes keys and values jointly across the full prefix in a single kernel pass; cache-ON computes them token-by-token and writes each result to the KV cache, producing a structurally different floating-point reduction order that the BF16 result confirms is a mantissa-precision mechanism rather than an FP16-specific exponent-range artifact (Section 5.7).

GQA amplification and decision boundaries. Multi-Query Attention (MQA; Shazeer 2019) is the special case in which one KV head is shared across all query heads ($R = H$); GQA generalizes this to $1 \leq R < H$; MHA is the case $R = 1$. Under cache-ON, a single FP16 rounding error in a shared KV head is broadcast to all R query heads simultaneously, producing R *correlated* rather than independent errors and amplifying divergence relative to MHA. The larger R is, the stronger the amplification: Mistral (GQA $R = 4$) exhibits systematically higher per-layer

drift than LLaMA-2 (MHA, $R = 1$). To our knowledge, this connection between GQA/MQA sharing ratio and FP16/BF16 cache divergence has not been previously established.

Hidden-state differences affect output only when competing tokens have similar log probabilities—that is, when the model is near a decision boundary. An early perturbation may switch a downstream decision not because it is large, but because the model is uncertain. This insight resolves the apparent contradiction between flip index and KL divergence (Section 5.4).

Mechanistic interpretability methods. Causal tracing (Meng et al., 2022) and path patching (Wang et al., 2022) intervene on residual-stream activations to identify causal variables, implicitly assuming the residual stream is the dominant causal site. We apply the same patching logic but target the KV cache state, finding that residual-stream patching fails (Section 5.5) while KV-tensor patching succeeds (Section 5.6).

LLM output consistency and determinism. Some work treats LLM output variation as stochastic (Zhang et al., 2023; Achiam et al., 2023) or due to prompt sensitivity (Zheng et al., 2023; Chen et al., 2023). We show cache-ON/cache-OFF divergence is deterministic, mechanistically traceable, and present even under greedy decoding. Yuan et al. (2025) show FP16/BF16 non-associativity leads to reproducibility failures *across* system configurations (batch size, GPU count, hardware). Our contribution is complementary: we demonstrate the effect on *identical* hardware with the same batch size, isolate the KV cache execution path as the source, and localize it through activation patching—aspects not examined by that work. Extended related work is in Appendix A.

3 Analysis Framework

Experimental logic. The five experiments form a single causal chain, each step answering a question the previous step leaves open: (1) *Does divergence exist, and is it structural?*—confirmed via token divergence and per-step KL under all model-strategy combinations, including greedy decoding to rule out sampling. (2) *How does it propagate?*—per-layer hidden-state drift shows whether the divergence is local or distributed and how GQA ratio and head dimension influence it. (3) *What causes it?*—FP32 re-runs test the non-associativity

hypothesis; elimination of divergence under FP32 identifies accumulation order as the sole causal driver. (4) *When does it matter behaviorally?*—decision-boundary proximity resolves the apparent flip-index/KL contradiction. (5) *Where does the divergence reside?*—activation patching of the residual stream tests causal localization; a null result localizes the causal variable to the KV cache state.

Three further analyses address possible confounds: (6) *How much of observed KL is pure numerical perturbation versus context compounding?*—teacher-forced conditional KL isolates the former. (7) *Is the effect FP16-specific, and can targeted precision changes mitigate it?*—BF16 replication and a mixed-precision ablation. (8) *Does token-level divergence translate into practically meaningful outcome differences?*—final-answer re-convergence versus a natural-resampling control.

Decoding strategies. We implement a custom autoregressive loop rather than high-level generation APIs, enabling simultaneous full-vocabulary logit retrieval under both cache-ON and cache-OFF at every decode step. For stochastic strategies, both paths share the same random seed at each step, ensuring token differences reflect distributional divergence rather than sampling variance. We evaluate under: (i) **greedy** ($T = 0$), which removes the sampling step so any effect is purely due to execution-path differences; (ii) **top- k** ($k = 50$, $T = 0.7$); and (iii) **nucleus sampling** ($p = 0.95$, $T = 0.7$). All sequences are decoded to a maximum length of 128 tokens.

Divergence metrics. We quantify divergence using four complementary metrics:

- **Token divergence** is a binary indicator of whether the two paths produce different token sequences at any position. Universal divergence indicates a structural property rather than random variation.
- **KL divergence** measures the distributional difference at each decode step. Let p_t and q_t denote the softmax distributions over the full vocabulary from cache-ON and cache-OFF at step t : $\text{KL}(p_t \| q_t) = \sum_v p_t(v) \log \frac{p_t(v)}{q_t(v)}$ (computed in nats). A persistently high KL trajectory indicates that distributional differences accumulate rather than arise incidentally. Note: the free-running metric conflates the pure numerical perturbation with divergence compounded by the two paths subsequently conditioning on different token histories; a conditional, teacher-

forced variant introduced in Step 6 isolates the former.

- **JS divergence** is the symmetric Jensen–Shannon divergence $\text{JS}(p_t \| q_t) \in [0, 1]$, confirming the KL pattern is not an artifact of KL’s sensitivity to low-probability vocabulary regions.
- **Flip index** is the decode step at which cache-ON and cache-OFF first produce different token predictions. It captures *when* divergence becomes behaviorally manifest rather than how large it is; its relationship with KL is analyzed in the decision-boundary analysis.

Statistical analysis. Each analysis uses a test matched to its comparison structure: McNemar’s test for paired correctness outcomes; Mann–Whitney U for per-layer drift magnitudes; Pearson and Spearman correlations for flip-index/KL relationships; Welch’s t -test for early- vs. late-flip KL comparisons. Multiple comparisons are corrected via Bonferroni and BH-FDR at $\alpha = 0.05$; bootstrap 95% CIs are reported throughout ($B = 10,000$ resamples). Full justifications, assumptions, and power analyses are in Appendix B.

4 Experiments

4.1 Setup

Models. We study three open-weight autoregressive models: **LLaMA-2-7B** (Touvron et al., 2023) (MHA, $R = 1$, head dimension 128), **Mistral-7B-v0.3** (Jiang et al., 2023) (GQA, $R = 4$, head dimension 128), and **Gemma-2-2B** (Team et al., 2024) (GQA, $R = 2$, head dimension 256, sliding-window attention). These models sample across attention type, GQA sharing ratio, head dimension, and parameter count. All models are loaded in FP16 via `HuggingFaceAutoModelForCausalLM.from_pretrained` with `FlashAttention` disabled unless specified; divergence was confirmed under both conditions.

Dataset. We evaluate on GSM8K (Cobbe et al., 2021), sampling 700 examples without replacement (`SAMPLE_SEED=42`) from the 1,319-example test set. Each example is evaluated under five different sampling seeds (0–4), for 3,500 runs per model-strategy combination and 31,500 runs across all nine combinations. We note that absolute accuracy values are low under our protocol (`max_new_tokens=128` may truncate some generations before completion; our few-

shot prompt format can cause models to mimic in-context exemplars). Accuracy differences should be interpreted as indicators of execution-path bias, not measures of absolute reasoning ability. To determine whether this divergence has consequences at more realistic generation lengths, we separately extend the evaluation to 400-token generations and to an instruction-tuned model (Mistral-7B-Instruct-v0.3) using its native chat template (Section 4.3); token divergence remains at 100% in both extensions.

We have 16 pairwise comparisons with achieved power ≥ 0.97 at $n = 3,500$; two underpowered comparisons (LLaMA-2 greedy vs. top- p and Gemma top- k vs. top- p) are treated as exploratory and excluded from confirmatory claims.

System environment. All experiments are conducted on a single NVIDIA H100 NVL GPU. Full hardware and software specifications are in Appendix C.

4.2 Experiment Design

Behavioral characterization. For each of the 31,500 run pairs, we record token divergence rate, KL divergence, flip index, and task accuracy. Each example is classified as always-, sometimes-, or never-diverging across five seeds, establishing whether divergence is a property of the example or the seed.

Layer drift analysis. For a subset of high-divergence examples, hidden-state representations are captured at each transformer layer under both paths during generation. Per-layer drift is reported at decode step 1, computed as the L2 norm of the hidden-state difference vector normalized by the cache-ON norm: $\|\mathbf{h}_{\text{on}} - \mathbf{h}_{\text{off}}\|/\|\mathbf{h}_{\text{on}}\|$. We report the normalized quantity as primary because raw L2 magnitude is confounded by residual-stream scale, which grows substantially and differently across models with depth (Appendix D.3). Per-head attention-weight KL divergence is also isolated to determine whether divergence appears in attention patterns or only in accumulated hidden-state values.

Root cause falsification. Four sub-experiments validate the FP16 accumulation mechanism (Appendix C), culminating in a full FP32 rerun ($n = 600$ per model). If FP16 accumulation order is the cause, FP32 must eliminate divergence entirely.

Decision boundary analysis. Run pairs are split into early-flip (bottom 33rd percentile of flip index)

and late-flip (top 33rd percentile) groups; Welch’s t -test tests whether early-flip examples have higher mean KL. Pearson correlations between flip index and mean KL, with and without boundary-case exclusions, assess whether decision-boundary proximity mediates the relationship.

Activation patching. For the $n = 600$ highest-KL examples per model (selected in decreasing order of mean KL from the full set of 700), cache-ON hidden states are injected into the cache-OFF forward pass at decode step 0—the first step at which the hidden states differ (prefill produces bitwise-identical states along both paths; $\|\mathbf{h}_{\text{on}} - \mathbf{h}_{\text{off}}\| = 0$ at all layers before decoding begins). Patching is applied one layer at a time and cumulatively across all layers. Recovery is measured as:

$$\text{pct_recovered} = \frac{\text{KL}_{\text{base}} - \text{KL}_{\text{patched}}}{\text{KL}_{\text{base}}} \times 100$$

where negative recovery confirms that patching increases divergence; all runs use MAX_STEPS=32.

4.3 Extended Validation Experiments

Teacher-forced conditional divergence. To isolate the pure numerical perturbation from divergence compounded by context drift (Step 6), we force cache-OFF to condition on cache-ON’s own free-running token sequence at every decode step, holding the input context identical throughout. Computed over $n = 300$ examples per model with bootstrap 95% CIs ($B = 10,000$).

Precision robustness. To test whether the effect is specific to FP16 (Step 7), we repeat the core divergence measurement under BF16 ($n = 50$ per model), and separately test a targeted mixed-precision ablation elevating only the attention computation (query/key/value projections through the softmax-weighted value sum) to FP32, immediately downcasting before the output projection, with the remainder of the forward pass remaining FP16.

Final-answer impact and reconvergence. Token-level divergence alone does not establish practical significance if trajectories frequently reconverge to the same final answer (Step 8). We measure whether diverging trajectories converge to the same final extracted answer and compare this reconvergence rate with a natural-resampling control: agreement between different random seeds under the *same* cache mode, representing ordinary sampling variance with no cache-mode difference, computed

under top- k /top- p sampling and excluding degenerate repetition-loop outputs.

5 Results

5.1 Behavioral Characterization

Table 1 summarizes divergence rate, mean KL, and task accuracy across all nine model-strategy conditions.

Token divergence is universal. Cache-ON and cache-OFF generate distinct sequences for all 31,500 run pairs (100% divergence rate, without exception). This result holds under greedy decoding—which eliminates the sampling step entirely—showing the effect is a structural property of FP16 cached inference, not sampling noise.

Execution-path bias is systematic. In 8 of the 9 conditions, cache-ON returns more correct answers than cache-OFF, and the differences reach Bonferroni-corrected significance (McNemar, $p < 0.05$). Only Gemma greedy is nonsignificant ($p = 0.43$), with the direction inverted: cache-OFF edges cache-ON (2.1% vs. 1.9%), suggesting Gemma’s greedy execution path does not reliably advantage either mode. The accuracy difference is an indicator of execution-path bias under our evaluation protocol, not a measure of absolute reasoning ability. The reliable trend in 8 of 9 conditions, rather than the absolute accuracy scores, is the primary finding.

KL divergence is large, stable, and model-specific. Mean KL varies from 10.68 (LLaMA-2 greedy) to 12.83 (Gemma top- p), with narrow bootstrap 95% CIs confirming stability across 3,500 run pairs per condition. Gemma consistently shows the greatest divergence. Within-model KL variation across strategies is small for LLaMA-2 and Gemma but notable for Mistral (greedy: 11.46 vs. top- k : 10.70), consistent with Mistral’s 4:1 GQA ratio amplifying per-step accumulation errors more aggressively than LLaMA-2’s MHA structure. Mean KL > 10 across all conditions indicates near-complete redistribution of probability mass at every decode step; these are not small numerical perturbations. JS divergence (bounded in $[0, 1]$) mirrors the KL trend in all conditions, confirming the pattern is not an artifact of KL’s unbounded range.

Figure 1 shows the per-step KL trajectory, which persists across all 128 steps with no convergence toward zero, consistent with FP16 errors accumulating rather than cancelling. Cross-model heatmaps of divergence rate and mean KL are in Appendix

Table 1: Behavioral results across all nine conditions ($n = 3,500$ run pairs). Divergence rate is 100% throughout. [†]Bonferroni-significant McNemar ($p < 0.05$); *nonsignificant. KL 95% CI is bootstrap. Accuracy values indicate execution-path bias under the 128-token truncation protocol, not reasoning ability; these are not comparable to published GSM8K results (typical 40–65% for 7B models under chain-of-thought prompting).

Model	Strategy	Div.	Mean KL	KL 95% CI	Cache Acc.	No-Cache Acc.
LLaMA-2	Greedy	100%	10.68	[10.63, 10.74]	3.3%	2.0% [†]
LLaMA-2	Top- k	100%	10.70	[10.65, 10.74]	3.3%	1.2% [†]
LLaMA-2	Top- p	100%	10.79	[10.75, 10.83]	2.8%	1.7% [†]
Mistral	Greedy	100%	11.46	[11.40, 11.53]	6.6%	2.0% [†]
Mistral	Top- k	100%	10.70	[10.66, 10.76]	6.5%	1.7% [†]
Mistral	Top- p	100%	10.90	[10.85, 10.94]	5.0%	1.9% [†]
Gemma	Greedy	100%	12.66	[12.62, 12.70]	1.9%	2.1%*
Gemma	Top- k	100%	12.80	[12.76, 12.84]	3.9%	1.1% [†]
Gemma	Top- p	100%	12.83	[12.79, 12.86]	3.3%	1.3% [†]

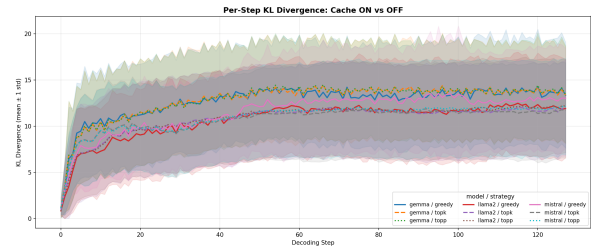


Figure 1: Mean KL divergence at each decode step across all models and strategies. Divergence persists across all 128 steps without convergence; Gemma shows the highest divergence, consistent with its GQA architecture and 256-dimensional heads.

Figure 6.

5.2 Layer Drift Analysis

Having established universal, systematic divergence, we ask how it propagates through the network. Figure 2 shows the per-layer hidden-state drift.

LLaMA-2 and Mistral amplify sharply at layer 1. LLaMA-2’s normalized drift jumps from 3.45 at layer 0 to 361.96 at layer 1 ($105\times$ amplification). Mistral’s increases from 25.76 to 902.36 ($35\times$), consistent with its 4:1 GQA ratio broadcasting a single FP16 error across four query heads simultaneously, producing correlated rather than independent errors. The raw L2 ranking reverses (LLaMA-2 922.29 vs. Mistral 336.62) because residual-stream scale grows differently across models with depth; the full raw-vs.-normalized comparison is in Appendix D.3.

Gemma shows a structurally distinct flat profile. Unlike LLaMA-2 and Mistral, Gemma does not experience a layer-1 amplification spike; its drift ranges from 1.0 to 1.4 across all 26 layers.

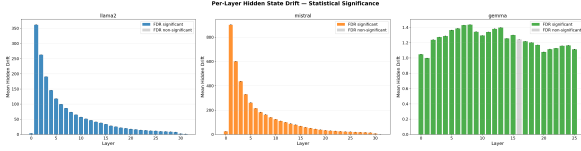


Figure 2: Per-layer hidden-state drift (normalized L2) at decode step 1. LLaMA-2 and Mistral show sharp amplification at layer 1 (105 $\times$ and 35 $\times$); Gemma shows a flat profile across all 26 layers.

Although Gemma has the lowest per-layer drift values, it has the largest total KL divergence in the behavioral characterization, suggesting its divergence is distributed across all layers rather than concentrated at a single amplification point. For Gemma, 25 of 26 layers are significant (BH-FDR); the exception is layer 16 ($p = 0.877$). Gemma’s flat profile is consistent with its larger head dimension (256 vs. 128); its sliding-window attention restricts each layer to local context rather than the full sequence, limiting global error propagation and producing gradual, uniform buildup of divergence across all layers rather than sharp amplification at a single point. 89 of 90 layer-model combinations are significant (Mann–Whitney U, BH-FDR, $\alpha = 0.05$).

Per-head attention-weight KL divergence is also isolated to confirm whether divergence appears in attention patterns or only in accumulated hidden-state values; results are consistent with the layer-level findings. Aggregate drift across all decoding strategies is in Appendix Figure 10; the absolute-vs.-normalized comparison is in Figure 8 and Table 9; per-model decision-boundary breakdowns are in Figures 11, 12, and 13.

5.3 Root Cause Falsification

FP32 eliminates divergence. Under FP32, KL divergence drops to a noise floor of $\approx 2.5 \times 10^{-8}$ in all three models (Figure 3, Table 2), and the flip rate is precisely 0.0% for all examples in all models. The FP16-to-FP32 KL ratio is $\sim 4 \times 10^8$ (more than eight orders of magnitude), identifying FP16 non-associativity as the dominant and sufficient cause. The divergence does not lie in the mechanism of the KV cache itself, but in the floating-point format chosen to implement it.

Accumulation error is architecture-determined. The average relative FP16 accumulation error is 0.036%; it remains flat across the sequence lengths tested (16–128 tokens) and is identical for all three models. This reflects

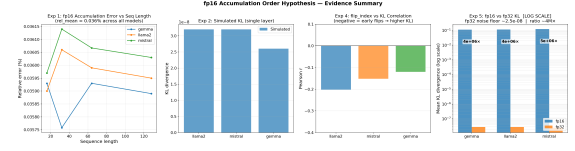


Figure 3: FP16 vs. FP32 KL divergence ($n = 600$ per model). FP32 collapses KL to a noise floor of $\approx 2.5 \times 10^{-8}$ and reduces flip rate to exactly 0.0%, isolating FP16 non-associativity as the sole cause. Accumulation error is flat across sequence lengths ($\approx 0.036\%$), confirming it is architecture-determined.

Table 2: FP16 vs. FP32 KL divergence and flip rate ($n = 600$, 32 decode steps). FP32 flip rate is exactly 0.0% across all models. FP16 KL values are lower than Table 1 due to shorter generation length (32 vs. 128 steps); the relevant quantity is the FP16-to-FP32 ratio.

Model	FP16 KL	FP32 KL	FP16 Flip	FP32 Flip
LLaMA-2	0.113	2.80×10^{-8}	1.15%	0.0%
Mistral	0.122	2.50×10^{-8}	1.14%	0.0%
Gemma	0.112	2.78×10^{-8}	1.03%	0.0%

the fact that per-step error magnitude is determined by the attention head dimension (d_{head}), not by sequence length, meaning divergence can be predicted from architecture alone without knowledge of the input.¹

5.4 Decision Boundary Analysis

Root-cause falsification confirms FP16 non-associativity as the cause, but an apparent contradiction remains: examples that flip *earlier* have *higher* KL divergence, seemingly contradicting the accumulation hypothesis (which predicts greater accumulated divergence should cause *later* flips). Figure 4 and Table 3 present four complementary views of this relationship and its resolution.

The correlation is consistently negative. Pearson correlation between flip index and mean KL is negative for all three models: $r = -0.202$ (LLaMA-2, $p < 10^{-96}$), $r = -0.152$ (Mistral, $p < 10^{-55}$), $r = -0.120$ (Gemma, $p < 10^{-34}$).

Decision-boundary proximity resolves the contradiction. KL is higher for early-flip examples than for late-flip examples in 8 of 9 conditions

¹Note on causal attribution: the falsification identifies the intersection of {FP16 arithmetic} $\cap$ {different accumulation order} as the causal agent. These two factors cannot be independently varied because the accumulation-order difference is a structural consequence of the KV cache mechanism, not a free parameter. An FP16 baseline with identical accumulation order in both paths cannot be constructed without eliminating the cache itself. The current experimental design is the strongest feasible test of the joint cause.

Table 3: Pearson correlation between flip index and mean KL ($n = 10,500$ per model); all correlations significant at $p < 10^{-34}$. GQA models show strengthened correlation after removing boundary cases, confirming decision-boundary proximity.

Model	Full r	Excl. flip = 1	Excl. flip ≤ 3	Trend
LLaMA-2	-0.202	-0.189	-0.115	weakens
Mistral	-0.152	-0.264	-0.333	strengthens
Gemma	-0.120	-0.177	-0.213	strengthens

(t -test, $p < 0.05$), with KL ratios ranging from 1.006 to 1.043. Examples close to the decision boundary have small logit margins and flip at the first decoding step because any perturbation can cross the threshold; these same examples have high KL because uncertainty spreads probability mass across tokens. Early flips and high KL are joint consequences of *uncertainty*, not causally linked to each other. The median flip index of 0 across all models (Figure 4, bottom-right) is consistent with this account: a single prefill-stage FP16 error suffices to flip token 1 near the decision boundary, while accumulation explains why KL remains persistently high across all 128 steps (Figure 1) rather than decaying after the first divergent token.

GQA models show a strengthening pattern. Excluding flip indices ≤ 3 makes the correlation more negative for Mistral ($-0.152 \rightarrow -0.333$) and Gemma ($-0.120 \rightarrow -0.213$), ruling out boundary-case outliers as drivers. The correlation weakens for LLaMA-2 ($-0.202 \rightarrow -0.115$), consistent with MHA-vs.-GQA differences in how boundary uncertainty propagates across decode steps. The cross-model flip-index/KL scatter is in Appendix Figure 7; per-model breakdowns are in Figures 11, 12, and 13.

5.5 Causal Localization

If divergence resides in the residual stream, patching cache-ON hidden states into cache-OFF’s forward pass at decode step 0 should recover the cache-free trajectory. If it does not, the divergence is consistent with localization in the KV cache state.

Residual-stream patching cannot recover KV cache divergence. As shown in Figure 5, single-layer patching results in a net median recovery of -0.019% across all layers and models (mean -1.21%). The high variance of individual-layer recoveries—LLaMA-2 ranges from -107% to $+56\%$ —indicates that single-layer patches perturb rather than correct the divergence trajectory. All

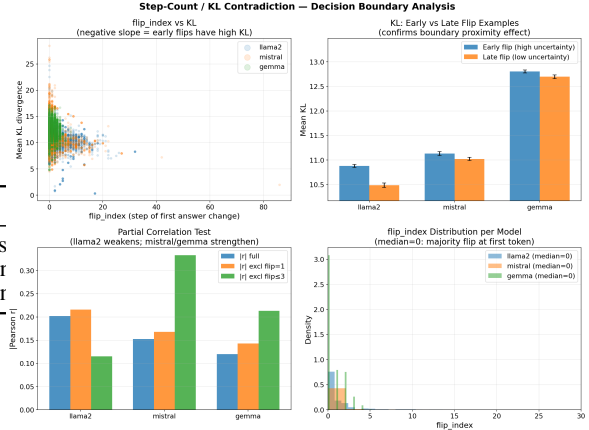


Figure 4: Four views of the flip index–KL relationship. Early-flip examples have higher KL (top row), resolved by decision-boundary proximity: both are effects of model uncertainty. Partial correlation strengthens in GQA models after removing boundary cases (bottom-left); the median flip index is 0 for all models (bottom-right).

three models have negative mean recovery when all layers are patched simultaneously: LLaMA-2, -5.34% ; Mistral, -2.56% ; Gemma, -2.88% . Negative recovery is not a failure; it is a mechanistically informative result: injecting cache-ON hidden states into the cache-OFF forward pass creates a cross-path inconsistency—the patched residual stream reflects a computation history shaped by cache-ON KV tensors, but all subsequent attention operations read from cache-OFF KV tensors. This hybrid state never occurred in either execution path and is therefore *further* from the cache-OFF trajectory than the unpatched state itself. The amplification of divergence under full-layer patching is direct evidence that the residual stream and KV cache are tightly coupled: correcting one without correcting the other increases divergence rather than reducing it. The full per-layer and cumulative recovery panel for all three models is in Appendix Figure 16.

Gemma’s partial recovery is architecturally explained. Its single-layer median recovery is $+5.75\%$ (range: 5.73% – 5.85% across all 26 layers), compared with $+0.08\%$ for LLaMA-2 and -8.83% for Mistral. Gemma’s flat drift profile—gradual, uniform accumulation with no single dominant spike—allows single-layer patches to partially redirect the trajectory, whereas the layer-1 spikes in LLaMA-2 and Mistral ($105\times$ and $35\times$) make single-layer interventions ineffective.

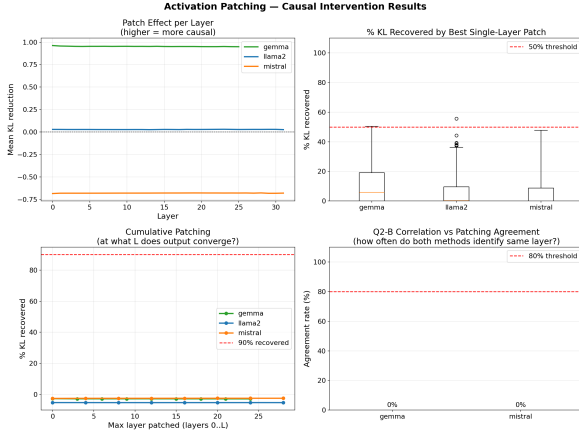


Figure 5: Activation-patching results. Single-layer median recovery is -0.019% across all layers and models; cumulative full-layer patching yields negative mean recovery in all three models, localizing the causal variable to the KV cache state.

5.6 Direct KV-Cache Tensor Patching

Residual-stream patching establishes the KV cache locus by elimination. This section closes the gap with a direct positive test.

Protocol. For $n = 50$ examples per model, cache-ON’s persistent key/value tensors are injected into cache-OFF’s cache at decode step 0, and the resulting output is re-evaluated. Concretely: (a) K and V tensors are patched *jointly* at every transformer layer; query projections, attention-weight computation, MLP, and all layer norms are left entirely untouched. (b) Patching is applied both individually (one layer in isolation) and cumulatively (layers 0 through ℓ , for increasing ℓ). (c) The patch is applied only at decode step 0 and measured at the single decode-step-0 output logit; recovery of 100% means the patched run’s output distribution matches cache-ON’s at that step—it does not mean identical full generated sequences. (d) *Compatibility construction*: because cache-ON’s cache spans the full prompt while cache-OFF’s chunked-recompute cache (window size 64 tokens, used throughout this paper) spans only the last 64 tokens, cache-ON’s tensors are sliced to the same 64-token relative window before patching, isolating the FP16 accumulation-order effect rather than conflating it with a context-window mismatch. (e) Only the cache-ON-into-cache-OFF direction was tested; the symmetric reverse direction is noted as a limitation.

Full-layer patch as a built-in self-check. Because cache-ON’s tensors are compatible with cache-OFF’s cache by construction and are sliced to an

Table 4: Direct KV-cache patching results ($n = 50$ per model). Layer-1 single-layer recovery is the % of baseline divergence closed by patching only layer 1 in isolation at decode step 0. Full-patch recovery is 100% by construction (see text).

Model	Mean KL_{base}	Layer-1 single-layer	Full-patch
LLaMA-2 (MHA)	7.12	39.18%	100.00%
Mistral (GQA 4:1)	8.45	0.29%	100.00%
Gemma (flat drift)	3.12	4.88%	100.00%

identical relative window, full-layer patching is *guaranteed to recover cache-ON’s decode-step-0 output exactly, independent of whether the KV-cache-locus hypothesis is correct*. This is a self-consistency check confirming the implementation is correct, not evidence on its own. The informative content is therefore (i) that residual-stream patching *fails* to recover (Section 5.5), which is a real test that could have failed, and (ii) the per-layer, per-model breakdown of which layers causally matter (Table 4).

Per-layer causal importance dissociates from passive drift. LLaMA-2 shows a single dominant causal layer: patching layer 1 alone recovers 39.18% of divergence, directly mirroring its layer-1 drift amplification. Mistral shows the opposite pattern despite having the *larger* passive drift amplification at layer 1: patching layer 1 alone recovers almost nothing (0.29%), while recovery builds gradually across layers with a notable nonlinearity—layers 0–11 patched jointly recover only 28.18%, but adding layer 12 (whose own single-layer recovery is negligible, -0.11%) increases cumulative recovery to 48.27%, indicating that layer 12’s correction matters only once earlier layers are already patched. Gemma shows high layer-to-layer volatility (single-layer recovery ranging from -32.58% to $+26.90\%$ across 26 layers), consistent with its flat, distributed drift profile: with no single dominant layer, patching any one layer can partially correct or locally worsen the divergence relative to the many unpatched layers. The key finding: a layer’s *passive* drift magnitude and its *active* causal importance to recovery are dissociable—Mistral’s layer 1 is the largest passive amplifier but among the least causally important layers to correct.

5.7 Extended Validation Results

Teacher-forced conditional divergence. Table 5 reports free-running versus teacher-forced KL ($n = 300$ per model, bootstrap 95% CI). The majority (80–93%) of the free-running metric’s magnitude

Table 5: Free-running vs. teacher-forced KL divergence ($n = 300$ per model, bootstrap 95% CI, $B = 10,000$).

Model	Free-running KL	Teacher-forced KL	% from context
LLaMA-2	8.24 [8.01, 8.46]	0.65 [0.61, 0.70]	91.5% [90.6, 92.2]
Mistral	9.05 [8.86, 9.24]	0.62 [0.59, 0.66]	92.9% [92.5, 93.3]
Gemma	10.14 [9.94, 10.35]	1.94 [1.85, 2.04]	80.5% [79.6, 81.4]

Table 6: FP16 vs. BF16 divergence severity ($n = 50$ per model).

Model	FP16 flip rate / KL	BF16 flip rate / KL
LLaMA-2	0.879 / 8.39	0.873 / 8.26
Mistral	0.920 / 9.14	0.908 / 9.05
Gemma	0.878 / 9.72	0.891 / 9.88

reflects context-compounding rather than the pure numerical effect. The teacher-forced KL remains clearly non-zero and orders of magnitude above the FP32 floor (2.5×10^{-8}), confirming the underlying numerical perturbation is real and measurable; it is distinct from, and substantially smaller than, its downstream compounding effect in free-running generation.

Precision robustness. Table 6 shows divergence severity under BF16 is statistically indistinguishable from FP16 across all three models, consistent with a mantissa-precision mechanism (BF16’s 7-bit mantissa is shorter than FP16’s 10-bit) rather than one specific to FP16’s exponent range.

The mixed-precision ablation (Table 7) elevates only the attention computation (query/key/value projections through the softmax-weighted value sum) to FP32, immediately downcasting before the output projection, with the remainder of the forward pass remaining FP16. Results are identical to three decimal places: attention-only precision elevation provides no measurable mitigation, in contrast to full-model FP32 (0.0% flip rate, Table 2). The protective effect of elevated precision requires it to be sustained across the *entire* forward pass, not in the attention mechanism alone. The full FP16-vs.-FP32 KL trajectory panel is in Appendix Figure 14; the condensed noise-floor view is in Figure 15.

Final-answer impact and reconvergence. Among token-divergent examples, 98.4% produce a *different final extracted answer* between cache-ON and cache-OFF; only 1.6% recover to an identical final answer despite intermediate token-level divergence. To rule out generally low task accuracy as an explanation, we compare with a natural-resampling control:

Table 7: Mixed-precision (attention-only FP32) ablation ($n = 50$ per model).

Model	FP16 baseline flip rate	Attention-only FP32 flip rate
LLaMA-2	0.867	0.867
Mistral	0.920	0.920
Gemma	0.878	0.878

Table 8: Cross-cache-mode recovery vs. natural resampling agreement.

Model	Cross-cache-mode recovery	Natural resampling agreement
LLaMA-2	2.36%	3.55%
Mistral	1.86%	4.68%

agreement between two different random seeds *under the same cache mode* (ordinary sampling variance, no cache-mode difference), computed under top- k /top- p sampling and excluding degenerate repetition-loop outputs from both quantities (Table 8). Cross-cache-mode recovery is lower than natural-resampling agreement in both LLaMA-2 and Mistral, confirming that cache-mode divergence suppresses reconvergence beyond what ordinary sampling variance alone would produce. Gemma showed a length-distribution artifact in this comparison—its cache-OFF path produces systematically shorter answers under top- k /top- p —and is excluded pending further investigation.

6 Discussion and Practical Guidance

A deeper mechanistic account underlying these recommendations is in Appendix D; limitations of the current study are in Appendix D.6; and concrete directions for future work are in Appendix E.

When this matters. Systems requiring bitwise or output-level reproducibility between cached and non-cached inference paths should not assume cache-ON/cache-OFF equivalence under FP16 or BF16. This includes validation pipelines that compare serving output against a reference implementation, deterministic replay used for debugging, and any context where reproducible model behavior is required for audit purposes. The reconvergence analysis confirms this is a practical concern: 98.4% of token-divergent examples produce a different final answer, and this effect exceeds what ordinary sampling variance alone would produce.

Detection. A lightweight diagnostic requires no statistical machinery. Running a small sample of representative inputs under both cache-ON and cache-OFF greedy decoding and checking for token-

sequence agreement reveals divergence in effectively all cases for the architectures we tested.

Mitigation trade-offs. Full FP32 inference eliminates the divergence (flip rate 0.0%) but incurs the full compute and memory cost, which may be prohibitive at production scale. Our mixed-precision ablation shows that partial-precision strategies confined to the attention computation are *not* sufficient—elevating only query/key/value projections and attention score computation to FP32 produced no measurable reduction, identical to three decimal places. Practitioners should not expect attention-only precision elevation to resolve this without independently validating end-to-end behavior. A full sweep of partial-precision strategies—elevating specific layers, or the KV cache write path specifically—remains a valuable direction for future work.

Precision migration. Practitioners considering a migration from FP16 to BF16 for other reasons (training stability, hardware support) should not expect this migration to incidentally resolve cache-ON/cache-OFF divergence. Our results show the effect persists at statistically indistinguishable severity under BF16, consistent with a mantissa-precision mechanism common to both formats.

Architecture-dependent severity. Our layer-wise analysis indicates GQA-based architectures may show different divergence propagation patterns than MHA-based architectures. Practitioners using GQA models should not assume mitigation strategies validated on MHA models transfer directly, and vice versa.

Interpreting divergence metrics. We caution practitioners building their own diagnostic tooling against naive free-running divergence metrics. These conflate the pure numerical perturbation with divergence compounded by the two paths subsequently conditioning on different token histories. Our teacher-forced conditional measurement shows this compounding effect can account for the large majority (80–93%) of a naive metric’s magnitude. Tooling intended to isolate the underlying numerical effect should adopt a similar conditional measurement.

7 Conclusion

The near-universal deployment of KV caching rests on an assumption rarely tested directly by system designers: that cached inference is numerically equivalent to cache-free inference. We have shown

this assumption fails under standard FP16 arithmetic, where differences in floating-point accumulation order between cache-ON and cache-OFF execution paths produce deterministic, reproducible divergence in decoded token sequences—for every model, strategy, and input we examined, without exception. The cause is FP16 non-associativity, confirmed by a clean falsification: switching to FP32 reduces divergence by more than eight orders of magnitude and eliminates token flips entirely. The divergence is carried forward by the KV cache state rather than the residual stream, propagates in architecturally predictable ways determined by attention variant and head dimension, and produces behavioral impact specifically when the model is near a decision boundary.

Beyond establishing the phenomenon, we characterized its boundaries and practical consequences. A conditional, teacher-forced measurement isolates the pure numerical perturbation from context-compounding divergence. The effect persists at comparable severity under BF16, ruling out FP16’s narrower exponent range as the explanation. A targeted mixed-precision ablation shows that elevating only the attention computation to FP32 is insufficient—mitigation requires precision sustained across the full forward pass. Finally, 98.4% of token-divergent examples produce different final answers, and this reconvergence failure exceeds what ordinary sampling variance alone would produce.

As GQA and MQA become dominant in production systems, the amplification effect we describe will widen the gap between cached and cache-free inference as models scale, making the correctness of cached inference a property that will matter more, not less, going forward. We provide practical guidance in Section 6 and identify direct KV-cache tensor patching and a broader sweep of partial-precision mitigation strategies as concrete directions for future work (Appendix E).

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thors’ own.

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A Extended Related Work

A.1 Numerical Precision in Neural Networks

Mixed-precision training (Micikevicius et al., 2017) showed that FP16 arithmetic with loss scaling enables training speedups without accuracy loss, while Gupta et al. (2015) showed deep networks can tolerate limited-precision arithmetic given negligible accuracy degradation. Later work extended reduced precision to inference: post-training quantization (Frantar et al., 2022), 8-bit matrix multiplication (Dettmers et al., 2022), activation stabilization (Xiao et al., 2023a), and activation-aware weight quantization (Lin et al., 2024). Earlier work also examined reduced precision and quantization

for efficiency (Micikevicius et al., 2017; Xiao et al., 2023b), but focused on training stability or model compression, not the interaction between accumulation order and KV cache execution path. No prior paper isolates the divergence caused specifically by FP16 non-associativity on deterministic cached inference.

A.2 KV Cache Optimization and Attention Efficiency

FlashAttention (Dao et al., 2022) and its successor (Dao, 2023) introduced IO-efficient exact-attention kernels that increase throughput by an order of magnitude; both implicitly rely on numerical equivalence across caching schemes. PagedAttention (Kwon et al., 2023) offered scalable KV cache management without analyzing differences in output correctness across execution paths. Multi-query attention (Shazeer, 2019) and grouped-query attention (Ainslie et al., 2023) reduce decoding costs by sharing keys and values across heads, but neither study considers the interaction between shared KV tensors and FP16 accumulation order. The original multi-head attention formulation (Vaswani et al., 2017) is the MHA baseline ($R = 1$) against which GQA amplification is measured throughout this paper. This is the first work to systematically measure divergence caused by execution-path variations in KV caching and demonstrate that the GQA ratio predicts divergence magnitude through correlated FP16 error propagation.

A.3 Mechanistic Interpretability

Causal tracing and ROME (Meng et al., 2022) introduced residual-stream interventions for discovering and editing facts in transformers, positing the residual stream as the dominant causal site. Path patching (Wang et al., 2022) isolated causal pathways for the indirect-object-identification task, while causal mediation analysis (Vig et al., 2020) formalized activation-level interventions for mechanistic attribution. Interpretable computational primitives such as induction heads were discovered under the assumption of a deterministic forward pass (Olsson et al., 2022). All these methods target the residual stream and implicitly assume divergence originates there. Our activation patching yields a null result: residual-stream patching cannot recover KV cache divergence even when all layers are patched simultaneously (Section 5.5). The KV cache state itself emerges as the causal variable, neglected by existing mechanistic work.

A.4 LLM Output Consistency and Reproducibility

Previous work investigated LLM output variability due to stochastic sampling and prompt sensitivity (Zheng et al., 2023), dataset-specific limitations (Bender et al., 2021), or systemic reproducibility issues across model versions and evaluation conditions (Chen et al., 2023; Achiam et al., 2023). Those studies treat variability as orthogonal to the model, dependent on stochastic effects or distribution shift. Our work is fundamentally different: we elicit reproducible, deterministic divergence solely from the KV cache execution path under FP16 inference, present even under greedy decoding and mechanistically explained by FP16 accumulation order, without the influence of any stochastic factor.

Most closely related, Yuan et al. (2025) show that FP16 and BF16 non-associativity can lead to reproducibility failures across system configurations (batch size, GPU count, hardware platform), and introduce LayerCast to mitigate this. Our findings differ in three ways: (1) we show divergence between cache-ON and cache-OFF paths on *identical* hardware with the same batch size; (2) we isolate the KV cache execution path as the source and localize it through activation patching (Section 5.5); (3) we analyze how the GQA sharing ratio predicts divergence magnitude, not explored in that work but highly relevant to contemporary LLM architectures.

B Statistical Analysis

Each test is selected to answer a particular inferential question in the argument.

Behavioral significance. McNemar’s test is applied to paired correctness results (cache-ON correct vs. cache-OFF correct on the same example). Pairing by example controls for prompt difficulty; the test asks whether one execution path systematically outperforms the other. Multiple comparisons are corrected using Bonferroni and Benjamini–Hochberg (BH-FDR) at $\alpha = 0.05$; bootstrap CIs ($B = 10,000$, 95% level) are given for mean KL and accuracy differences.

Layer drift significance. Mann–Whitney U tests are run on per-layer hidden-state drift magnitudes across 90 layer-model combinations, with BH-FDR correction. The nonparametric test is chosen because drift magnitudes are right-skewed.

Step-count correlation. Pearson correlations between flip index and mean KL are computed at $n = 10,500$ per model, with partial correlations excluding flip index = 1 and flip indices ≤ 3 to test whether the negative relationship holds without the most extreme boundary flips.

Power analysis. We have 16 pairwise comparisons with achieved power ≥ 0.97 at $n = 3,500$; two underpowered comparisons (LLaMA-2 greedy vs. top- p and Gemma top- k vs. top- p) are treated as exploratory and excluded from confirmatory claims.

C Experiment Environment and Sub-experiments

C.1 System Environment

All experiments are conducted on a single NVIDIA H100 NVL (95 GB) GPU, paired with an AMD EPYC 9254 24-core CPU and 250 GB of RAM. The system runs Ubuntu 22.04 with Python 3.12.3, PyTorch 2.5.1 (CUDA 12.1), and Hugging Face Transformers 4.53.0. For greedy decoding, we set `CUBLAS_WORKSPACE_CONFIG=:4096:8` and enable deterministic behavior via `torch.use_deterministic_algorithms(True)`, ensuring bitwise reproducibility across reruns on the same GPU.

C.2 Root Cause Sub-experiments 1–4

Experiment 1 quantifies the relative FP16 accumulation error of dot products versus sequence length ($\approx 0.036\%$, flat across all lengths and models, standard deviation $= 8.5 \times 10^{-7}$).

Experiment 2 simulates multilayer error propagation through N sequential attention and residual steps, showing how single-layer FP16 rounding errors compound across layers, consistent with the large KL divergence observed behaviorally.

Experiment 3 measures the FP16 rounding error in K and V projections between joint computation (cache-OFF) and token-by-token computation (cache-ON), confirming the error originates at *cache-write time* rather than downstream.

Experiment 4 correlates the flip index with mean KL divergence to assess whether decision-boundary proximity mediates the effect.

D Extended Mechanistic Discussion

Practical guidance is given in Section 6; this appendix provides the deeper mechanistic account underlying those recommendations.

D.1 A Unified Mechanistic Account

The five experiments form one unifying argument. Behavioral characterization shows that cache-ON and cache-OFF always diverge in FP16 across every run, model, strategy, and seed. Root-cause falsification identifies the cause: FP16 non-associativity causes the two paths to accumulate in different orders, and FP32 eliminates the divergence. Layer-drift analysis demonstrates how the initial error propagates—rapidly at layer 1 for LLaMA-2 and Mistral (Figure 2), broadly distributed in Gemma. Decision-boundary analysis shows *when* this matters behaviorally: not when divergence is large in magnitude, but when the model is near a decision boundary (Figure 4). Causal localization establishes *where* the error resides: patching the entire residual stream fails even when all layers are patched simultaneously (Figure 5), confirming the KV cache state carries the error forward.

This mechanism is deterministic, reproducible, architecturally predictable, and cannot be corrected by post-processing. It differs clearly from stochastic output fluctuations (Zheng et al., 2023; Chen et al., 2023) and from batch-size- and hardware-dependent nondeterminism (Yuan et al., 2025).

D.2 Architectural Predictability

Divergence can be predicted from attention architecture. The 0.036% average relative accumulation error is flat across sequence lengths and models, establishing that per-step error magnitude is determined by head dimension and attention structure, not input context. When combined with GQA, one FP16 error propagates to R query heads simultaneously, producing R correlated rather than independent errors. Within-model layer-1 amplification reaches $105\times$ for LLaMA-2 and $35\times$ for Mistral (Figure 2), consistent with GQA broadcasting the same error to all four query heads.

D.3 Raw vs. Normalized Layer-1 Drift

Table 9 and Figure 8 report both quantities for transparency. The normalized ranking (Mistral > LLaMA-2) is consistent with the GQA-amplification account; the raw ranking reverses (Figure 8), illustrating that raw magnitude alone is not a reliable basis for cross-model comparison. Figure 9 shows that the layer-1 amplification pattern persists at later decode steps.

Table 9: Layer-1 drift: raw vs. normalized (mean and median across examples).

Model	Raw L2 (mean)	Normalized (mean)	Normalized (median)
LLaMA-2-7B	922.29	361.96	360.62
Mistral-7B	336.62	902.36	897.34
Gemma-2-2B	74.09	1.00	1.03

D.4 The Localization Argument and Its Limits

Causal localization (Section 5.5) establishes that divergence does not reside in the residual stream; direct KV-tensor patching (Section 5.6) converts this elimination argument into positive confirmation, subject to stated scope (joint K/V patching, step-0 measurement, single direction). The results together establish a necessary condition—no residual-stream intervention recovers the divergence—and, within scope, a sufficient one at decode step 0: restoring the KV cache state does. Figure 16 shows the full per-layer and cumulative patching panel for all three models.

D.5 Implications

- (i) **Cache equivalence cannot be assumed.** All FP16 production systems leveraging KV caching yield results distinct from those of no-cache inference in 100% of cases.
- (ii) **Divergence direction is systematic.** Cache-ON yields higher accuracy than cache-OFF in 8 of 9 conditions (Section 5), indicating systematic execution-path bias rather than symmetric noise.
- (iii) **GQA amplifies the effect.** Such divergence will be more common as GQA and MQA become dominant in modern LLMs.
- (iv) **The effect is not specific to FP16.** BF16 reproduces the divergence at statistically indistinguishable severity (Table 6), consistent with a mantissa-precision mechanism common to both formats.

D.6 Limitations

- (i) **Scope.** Three models and one benchmark represent important architectural dimensions but not the full spectrum. We confirm 100% token divergence persists at `max_new_tokens=400` across all three base models and on Mistral-7B-Instruct-v0.3 with its native chat template. Full validation at generation lengths typical of extended reasoning traces (e.g., AIME- or

LiveCodeBench-style outputs) on reasoning-tuned models and additional benchmarks remains future work. Our root-cause analysis predicts the effect should persist or compound at such lengths—the per-step accumulation error is architecture-determined, so KL should accumulate monotonically—but this has not been directly confirmed at that scale.

(ii) **Evaluation protocol.** Absolute accuracy values are low due to `max_new_tokens=128` truncation and few-shot prompt format. Accuracy differences serve as indicators of execution-path bias, not reasoning performance. The core results—100% divergence and systematic direction—do not depend on absolute accuracy levels. We additionally verify that 98.4% of divergent examples produce different final answers (Table 8), exceeding the reconvergence failure rate under ordinary sampling variance alone.

(iii) **Direct KV-cache patching: scope, not absence.** The intervention patches K and V jointly (not separately), at decode step 0 only (not across full trajectories), and in one direction only (cache-ON into cache-OFF). Three concrete low-cost extensions remain open: separating K vs. V contributions; testing step-by-step patching across full trajectories; and running the symmetric reverse-direction check.

(iv) **Quantized inference.** This paper studies FP16 inference; integer-quantized models (INT8, INT4/GPTQ, AWQ) represent a distinct regime. Under INT8 KV quantization the FP16 rounding mechanism would be dominated by quantization noise. Whether INT8 KV caching introduces its own form of cache-ON/cache-OFF inequivalence through asymmetric quantization rounding is a natural extension.

E Future Work

The most proximal extensions to the direct KV-cache tensor patching result (Section 5.6) are: separating the causal contribution of K versus V tensors; testing whether step-by-step patching sustains trajectory-level agreement across a full generation; and running the symmetric reverse-direction patch as a robustness check. None are blocked on

tooling—the current implementation already supports arbitrary layer/step targeting.

Beyond the five-step causal chain, open questions include: formal quantification of the GQA amplification factor across a wider range of sharing ratios with matched parameter counts; validation at reasoning-model-scale generation lengths (potentially several thousand tokens); and characterization of the interplay between KV cache divergence and cache-compression schemes (quantization, eviction, low-rank methods), for which additional precision loss may compound the reported effects.

Adversarial-robustness-style experiments that inject pre-specified noise into KV tensors and measure flip rate as a function of noise amplitude would determine susceptibility thresholds and clarify how close to a decision boundary a model must be for divergence of a given magnitude to produce a behavioral flip (complementing the decision-boundary analysis of Section 5.4).

Partial-precision mitigation strategies beyond the attention-only ablation (Table 7) also merit investigation: elevating specific layers (e.g., layer 1 in LLaMA-2 given its 39% single-layer recovery, Table 4), or elevating precision specifically on the KV cache write path, may offer a better compute/quality trade-off than full FP32 inference.

F Additional Figures

Figures 6–17 provide supplementary visualizations for all five experimental steps. They are referenced from the relevant subsections of the main body where noted.

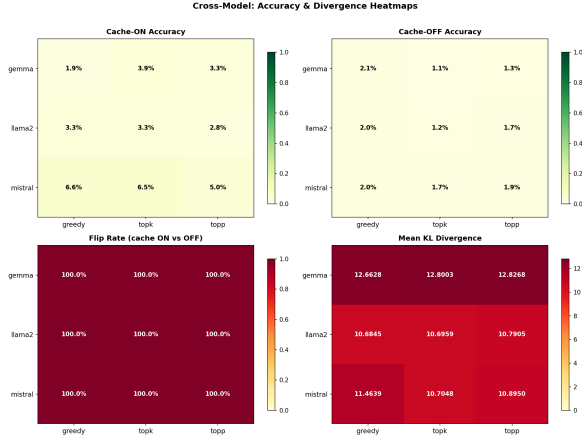


Figure 6: Cross-model heatmaps of token divergence rate and mean KL divergence across all nine model-strategy conditions. Complements Table 1: the heatmap view makes within-strategy and within-model variation immediately visible. Gemma’s consistently higher KL and Mistral’s cross-strategy variance are both apparent.

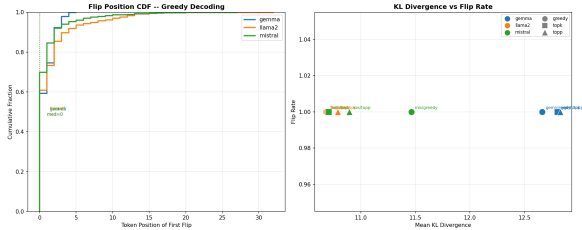


Figure 7: Cross-model scatter of flip index versus mean KL divergence. The negative correlation ($r = -0.120$ to -0.202 , Table 3) is visible across all three models; early-flip examples cluster at high KL, consistent with the decision-boundary account of Section 5.4.

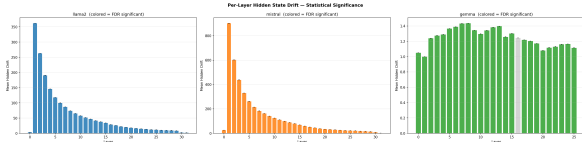


Figure 8: Per-layer hidden-state drift (absolute L2 norm) at decode step 1. Compare with Figure 2 (normalized, main body): the raw ranking reverses for LLaMA-2 and Mistral because residual-stream scale grows substantially and differently across models with depth (Table 9).

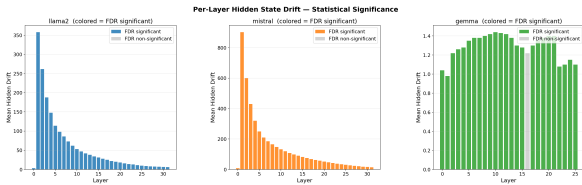


Figure 9: Per-layer drift at later decode steps (> 1), showing that the layer-1 amplification pattern observed at step 1 (Figure 2) persists across the full decoding trajectory.

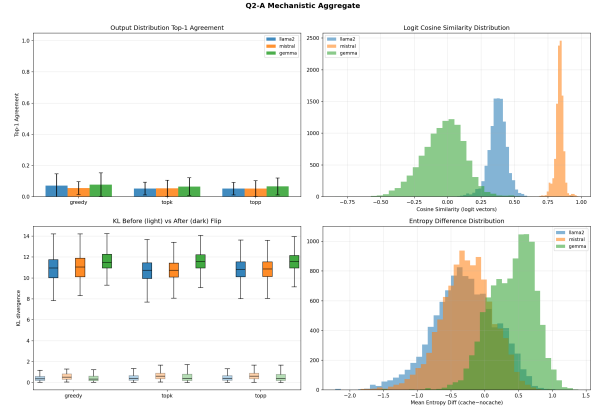


Figure 10: Aggregate layer drift across all models and decoding strategies. The per-layer amplification profile is consistent across greedy, top- k , and top- p decoding, confirming that the pattern is not specific to one decoding regime.

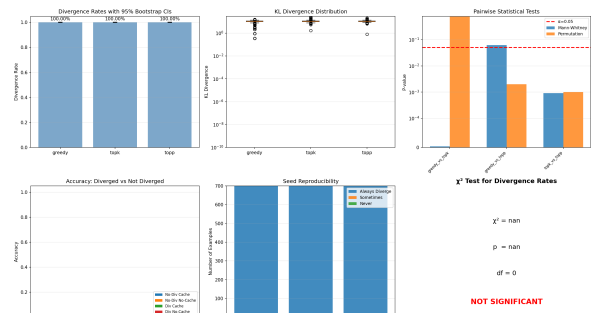


Figure 11: Decision boundary analysis for LLaMA-2: full distribution of flip index and mean KL for early- and late-flip groups. Complements the aggregate view in Figure 4: early-flip examples have higher KL, consistent with the decision-boundary proximity account (Section 5.4).

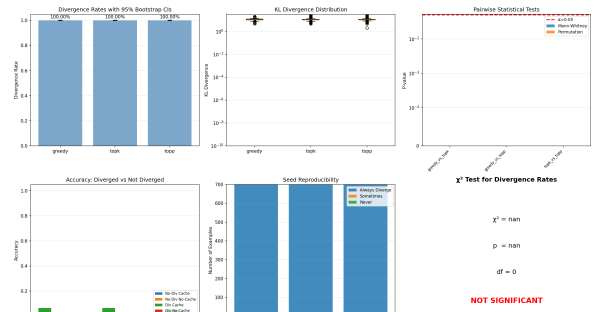


Figure 12: Decision boundary analysis for Mistral. The negative flip-index/KL correlation strengthens after removing boundary cases ($-0.152 \rightarrow -0.333$, Table 3), consistent with GQA amplification of boundary uncertainty.

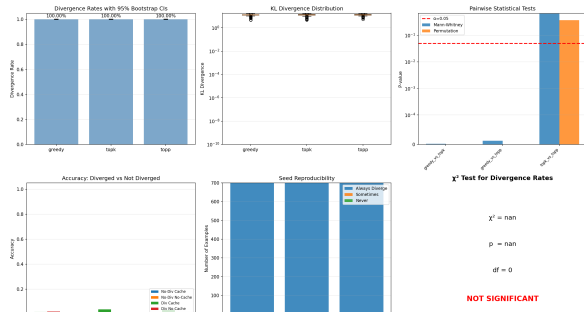


Figure 13: Decision boundary analysis for Gemma. Despite Gemma’s flat per-layer drift profile, the flip-index/KL correlation ($r = -0.120$) and early/late-flip KL difference follow the same pattern as LLaMA-2 and Mistral.

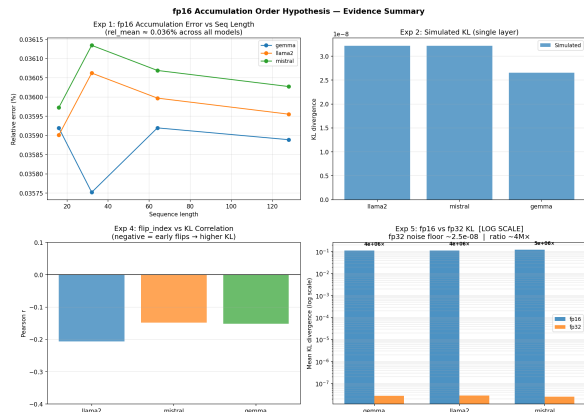


Figure 14: Full hypothesis summary panel: FP16 vs. FP32 KL divergence trajectories and flip rate comparison across all three models. Expands Figure 3 (main body) with per-step KL trajectories under both precisions.

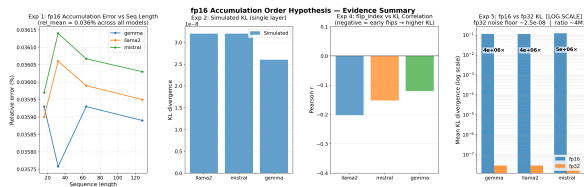


Figure 15: Condensed single-row hypothesis summary showing the FP32 noise floor for each model (Table 2). The $\approx 2.5 \times 10^{-8}$ floor is visually indistinguishable from zero at the FP16 scale.

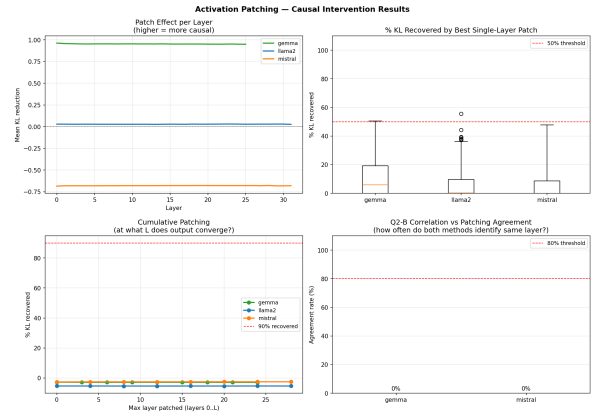


Figure 16: Full activation-patching results panel: per-layer recovery (left) and cumulative recovery (right) for all three models. Expands Figure 5 (main body) with the complete per-layer breakdown. LLaMA-2’s high variance (-107% to $+56\%$ at individual layers) and Gemma’s uniform partial recovery ($+5.75\%$ per layer) are visible.

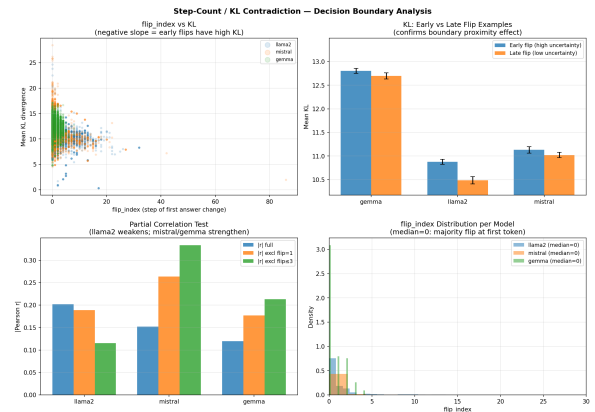


Figure 17: Alternative view of flip-index and step-count distribution, complementing Figure 4 (main body). Shows the full marginal distributions of flip index across models; the median of 0 is apparent for all three.